Hero Adjusted Plus–Minus in *Marvel Rivals*

Intention-to-treat estimates from 477,483 competitive matches

Season 9.5, PC ranked. Prepared 8 September 2026. Commit `31cfdff`.

This report estimates the value of each hero in *Marvel Rivals* using 477,483 competitive PC ranked matches from Season 9.5. The goal is to separate a hero's association with winning from the many other features of a match that affect the outcome, including map, player skill, team composition, and team-up effects. The estimates should be interpreted as adjusted plus–minus: how much the probability of winning changes when a team starts a particular hero instead of an average hero in the same role, holding the other observed features of the match fixed.

A first difficulty is that team composition matters independently of hero identity. Teams field different role structures, and the standard 2–2–2 structure of two Tanks, two Damage heroes, and two Supports, which 75% of teams use, wins 53.5% of its matches, while off-meta structures such as one Tank with three Damage heroes win 42.2% and two Tanks with three Supports win 40.4%. Whatever heroes fill them, unusual role structures lose more often. The model therefore controls for team composition directly: each composition shape, defined by the number of Tanks, Damage heroes, and Supports fielded, that occurs at least 100 times receives its own indicator, entered as the difference between the two teams, so that unusual role structures are absorbed by the composition controls rather than attributed to individual heroes.

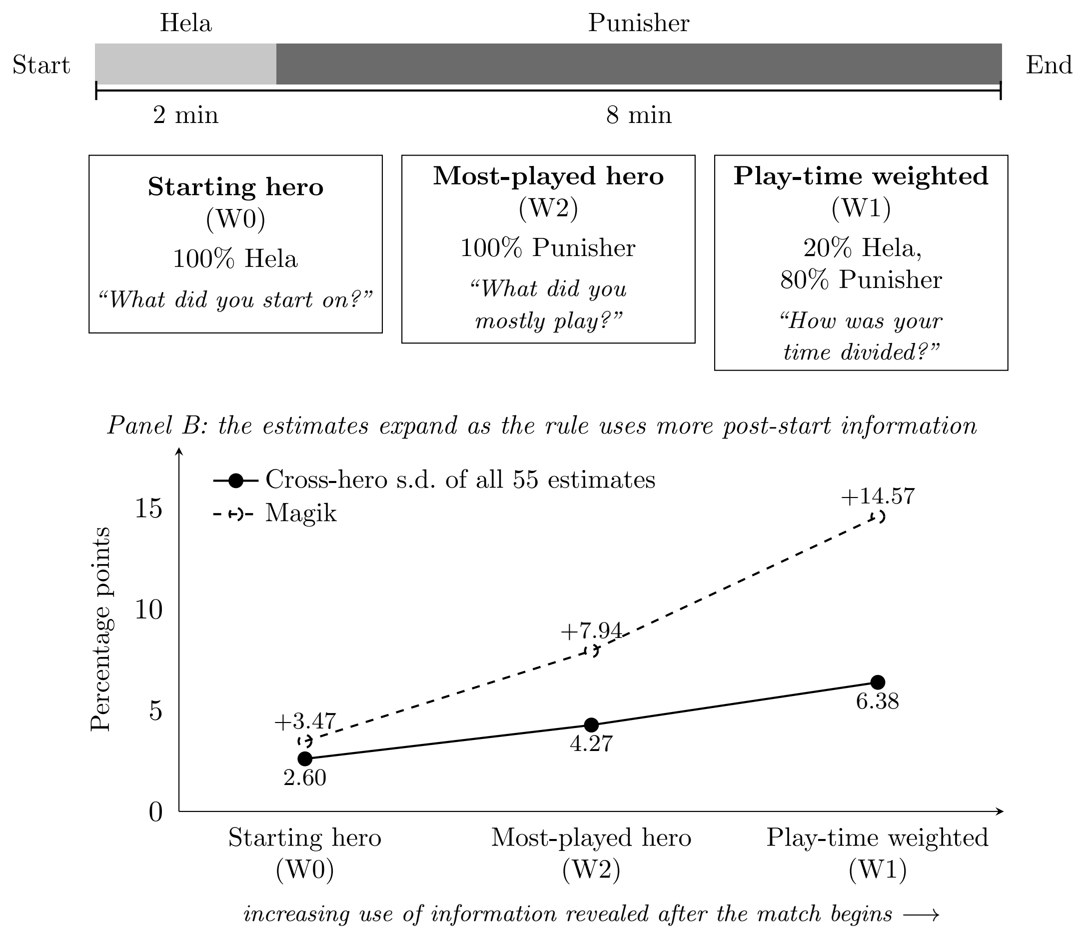
The unit of observation is a match, and the outcome is whether side 0 wins. Each hero enters the model as a signed contrast between the two teams: the hero's weight on side 0 minus its weight on side 1. Hero coefficients are estimated subject to a sum-to-zero restriction within each role, so every Tank is compared with the average Tank, every Damage hero with the average Damage hero, and every Support with the average Support. The regression also includes a separate intercept for each of the 16 maps, the pre-match rank-score differential between the teams, composition-shape indicators, and 100 team-up contrasts. The model is estimated using unpenalized logistic regression with HC1 heteroskedasticity-robust standard errors.

Hero attribution is based on the hero each player starts the match on (starting-hero attribution; W0 in the appendix tables). The API records heroes chronologically by first appearance, so the first entry provides an assignment that is fixed before the outcome of the match is known. This gives the estimates an intention-to-treat interpretation. If a player starts on a hero and switches thirty seconds later, the original hero still receives full attribution. That necessarily attenuates the estimated effect of heroes that are frequently abandoned, but it avoids allowing information generated during the match to determine the regressors. In this setting, the tendency for players to abandon a hero is also arguably part of the hero's practical value.

An apparently attractive alternative is to weight heroes by the amount of time they are played (play-time weighting; W1). Figure 1 illustrates the choice with a player who swaps once, together with a third rule, most-played-hero attribution (W2), which assigns the whole match to whichever hero the player used longest. The data show why play-time weighting is problematic. Moving from starting-hero attribution to increasingly outcome-dependent rules makes the estimated effects systematically larger and improves in-sample fit. Across the same set of matches, the cross-hero standard deviation of the estimates rises from 2.60 percentage points under starting-hero attribution to 4.27 under most-played-hero attribution and 6.38 under play-time weighting. For a single hero the path can be dramatic: Magik's estimate moves from +3.47 to +7.94 to +14.57 points across the three rules. The ranking of heroes remains fairly stable, with rank correlations of 0.937 and 0.890 against the starting-hero estimates, but the magnitudes expand as more post-start information enters the regressors. That pattern is exactly what we would expect if play-time weighting partly encodes what happened during the match. For that reason, starting-hero attribution is used for the headline estimates.

Figure 1: Attribution rules when a player swaps heroes, and what they do to the estimates

Panel A: one player, one swap (illustrative match)

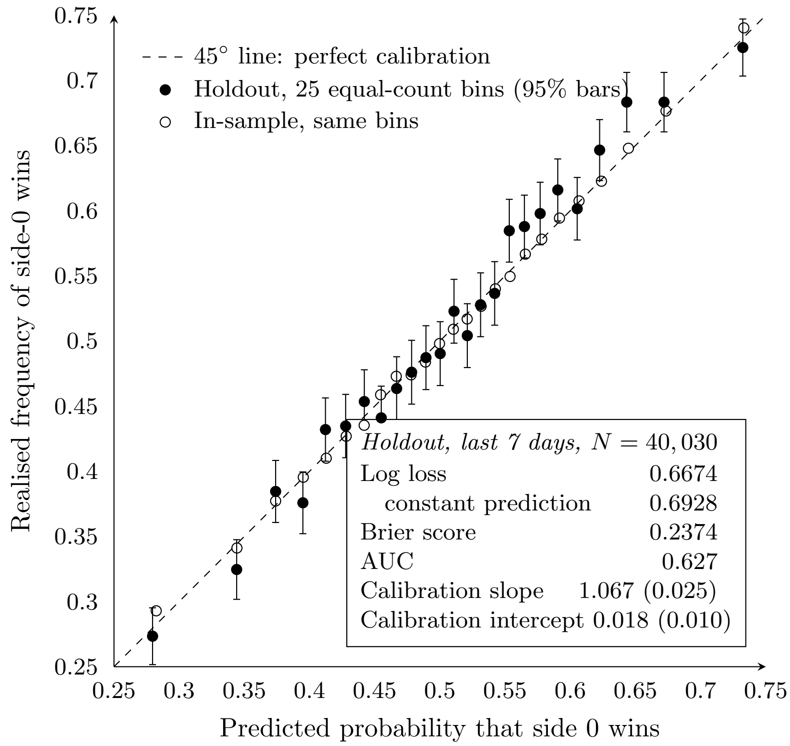


Notes. Panel A is a stylised match; the three rules differ only in how they treat a player who swaps. Panel B plots, for each rule, the cross-hero standard deviation of the 55 within-role estimates (filled circles), all fitted on the same 477,483 matches, and the estimate for Magik (open circles), the hero whose estimate moves most between the starting-hero and play-time-weighted rules. Appendix Table A3 lists every hero under every rule.

The within-role normalization is also important for identification. Cross-role comparisons cannot cleanly be interpreted as hero effects because the number of Tanks, Damage heroes, and Supports is itself a property of the team composition. Under a single global normalization, the hero block and the composition block share a nearly collinear direction corresponding to role counts. Imposing a separate sum-to-zero restriction within each role removes that direction from the hero coefficients entirely. Composition effects are therefore estimated by the composition controls, while hero coefficients measure variation among heroes occupying the same role. Re-estimating the model under this parameterization leaves the substantive results almost unchanged: the rank correlation with the earlier estimates is 0.9999 and the mean absolute change is only 0.010 percentage points.

The resulting estimates show substantial dispersion within every role (Table 1). Among Tanks, The Thing has the largest positive adjusted effect at 5.46 percentage points relative to the average Tank, followed by Magneto, Angela, Peni Parker, Hulk, and Thor. Among Damage heroes, Black Cat leads at 4.85 points, followed by Magik, Storm, Daredevil, and Mister Fantastic. Among Supports, Mantis stands out most strongly at 6.76 points, followed by Rocket Raccoon, Ultron, Adam Warlock, and Loki. These numbers are marginal effects evaluated at a balanced match, so an estimate of 5.46 means that replacing an average hero of the same role with The Thing at match start is associated with approximately a 5.46

Figure 2: Out-of-sample calibration of the headline model



Notes. The model is fitted on the 437,453 matches played before the final 7 days of the sample and evaluated on the 40,030 matches played during them; a random split would leak meta drift. Matches are sorted by predicted probability into 25 equal-count bins of about 1,600 matches; each filled circle is one bin's mean prediction against its realised win frequency, with a 95% binomial interval; open circles repeat the construction in-sample on all 477,483 matches. The calibration slope and intercept are from a logit of the realised outcome on the log-odds of the prediction (ideal: 1 and 0). A constant prediction at the training base rate gives the log loss shown for comparison. Predicted probabilities fall between 5% and 93%, and 85% of holdout matches are predicted between 35% and 65%: the model separates unbalanced matches well and, as it should, treats balanced ones as coin flips.

percentage-point increase in win probability, conditional on the included controls. 10 of the 55 heroes cannot be distinguished from their role average after false-discovery-rate correction.

Several specification checks support the interpretation of the model (Appendix Table A1). Randomly permuting hero assignments destroys 97.4% of the measured signal, indicating that the estimates are not simply being generated mechanically by the design matrix. The within-role restrictions hold to machine precision. Correcting the role-composition identification problem also raises the rank correlation between adjusted estimates and raw hero win rates from 0.671 to 0.801, while the within-role reparameterization itself leaves the coefficients essentially unchanged. Map-specific intercepts are retained because absolute camp-0 win rates vary from 49.42% to 52.52% across the 16 maps.

The model also predicts well out of sample. Fitted on every match except those from the final 7 days and evaluated on the 40,030 matches of that week, it is well calibrated (Figure 2): across 25 equal-count bins of predicted win probability, the realised win frequency tracks the 45° line, and the calibration slope is 1.067 (standard error 0.025), slightly above the ideal of one, so if anything the model is mildly under-confident out of sample. The holdout log loss is 0.6674, compared with 0.6928 for a constant prediction, and the area under the ROC curve is 0.627. Appendix Table A2 shows how much each block of the model contributes to that out-of-sample fit; adding the hero contrasts to a model that already holds map, rank score, and composition fixed lowers the holdout log loss from 0.6819 to 0.6698.

The team-up estimates (Appendix Table A4) capture an additional object: the value associated with fielding a designated pair beyond the sum of the two heroes' individual effects. For example, the Squirrel

Girl–Iron Man pair carries an estimated premium of 8.66 percentage points, Blade–Moon Knight 8.30 points, Rogue–Gambit 6.95 points, and Adam Warlock–Ultron 6.82 points. These coefficients should therefore not be read as the total strength of the pair. They measure only the incremental association with winning beyond what the two component heroes already contribute separately.

The results nevertheless have several important limitations. First, the exact magnitudes depend on the attribution rule more than the ordering does. Starting-hero attribution produces a cross-hero standard deviation of 2.60 percentage points, compared with 4.27 under most-played-hero attribution and 6.38 under play-time weighting, while rank correlations with the starting-hero estimates remain 0.937 and 0.890. The relative ordering of heroes is consequently more robust than the absolute size of the coefficients.

Second, these estimates are adjusted associations rather than causal treatment effects. Hero selection is endogenous. The rank-score control captures differences in general player skill, but it does not measure a player's hero-specific proficiency. A highly experienced one-trick is not equivalent to a randomly selected player assigned the same hero. Player-by-hero experience would provide a direct control for this source of selection, but that information is not available in the current specification. The model therefore reduces this confounding rather than eliminating it.

Third, the reported uncertainty is likely too optimistic. The confidence intervals use HC1 standard errors and treat matches as the primary observations, even though each match contains twelve players and the same player may appear in many matches. The resulting dependence structure is cross-classified and overlapping. A player-cluster bootstrap would better reflect this dependence, but the specified implementation would require roughly 656 hours of computation at the current sample size. The point estimates are unaffected by this issue, but the reported confidence intervals should be interpreted as narrower than fully dependence-robust intervals would be.

Finally, the composition controls are necessarily coarse in the extreme tail of the distribution, and the sample itself is not representative of the entire ranked population. 18 composition shapes receive separate indicators, while the remaining 9 rare shapes, together only 0.024% of team instances, are pooled into an "other" category. In addition, profile privacy rises sharply at high rank scores, reaching roughly 63% above 5,000. Because the crawl can expand only through public profiles, the highest-ranked portion of the player population is structurally under-sampled.

Taken together, the estimates are best viewed as a large-sample measure of how hero choice at the start of a match is associated with subsequent winning after accounting for observable differences in map, team strength, team structure, and designated team-up effects. They are most informative for comparing heroes within the same role, and the ranking of heroes is more credible than treating every estimated percentage-point difference as a literal causal effect.

Table 1: Hero adjusted plus–minus, relative to an average hero of the same role

	APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time − Starting		APM	S.E.	q	Starts	Pick (%)	Raw WR (%)	Play-time − Starting
Panel A: Tank (15 heroes)								*Panel B: Damage (27 heroes)*							
The Thing	5.46***	(0.204)	0.000	122,883	2.12	52.4	+1.75	Black Cat	4.85***	(0.260)	0.000	66,494	1.15	55.2	+6.77
Magneto	1.96***	(0.139)	0.000	400,170	6.90	50.9	+1.60	Magik	3.47***	(0.219)	0.000	153,326	2.65	56.8	+11.11
Angela	1.87***	(0.261)	0.000	45,696	0.79	51.8	+0.81	Storm	3.09***	(0.305)	0.000	45,132	0.78	55.3	+7.18
Peni Parker	1.51***	(0.328)	0.000	43,799	0.76	54.6	+5.90	Daredevil	2.92***	(0.217)	0.000	75,418	1.30	53.6	+5.55
Hulk	1.33***	(0.219)	0.000	81,337	1.40	52.3	+2.21	Mister Fantastic	2.52***	(0.351)	0.000	40,940	0.71	50.4	+1.43
Thor	1.29***	(0.219)	0.000	80,452	1.39	50.4	-0.64	Iron Man	1.34***	(0.279)	0.000	50,391	0.87	51.5	+1.76
The Hood	0.45**	(0.172)	0.012	404,062	6.97	52.8	+6.08	Iron Fist	1.32***	(0.224)	0.000	85,109	1.47	52.1	+3.19
Devil Dinosaur	-0.03	(0.231)	0.907	69,629	1.20	48.6	-2.21	Hela	1.14***	(0.199)	0.000	110,771	1.91	49.8	+0.08
Groot	-0.67**	(0.290)	0.027	58,919	1.02	49.9	-3.13	Winter Soldier	1.00***	(0.141)	0.000	228,817	3.95	49.3	-0.83
Venom	-0.93***	(0.219)	0.000	103,844	1.79	50.2	-0.41	Wolverine	0.82**	(0.317)	0.012	42,210	0.73	49.3	-0.61
Doctor Strange	-0.98***	(0.202)	0.000	135,405	2.34	48.4	-5.85	Spider-man	0.58***	(0.175)	0.001	142,393	2.46	50.9	+2.32
Captain America	-1.22***	(0.352)	0.001	33,294	0.57	50.4	-0.57	Moon Knight	0.10	(0.297)	0.792	54,845	0.95	49.0	-3.47
Emma Frost	-1.77***	(0.192)	0.000	167,701	2.89	46.9	-4.91	Black Widow	0.02	(0.200)	0.907	84,335	1.46	48.7	-1.26
Rogue	-3.41***	(0.344)	0.000	52,686	0.91	51.0	+1.64	Blade	-0.20	(0.317)	0.578	47,659	0.82	49.8	+1.28
Deadpool (Vanguard)	-4.86***	(0.252)	0.000	40,623	0.70	47.9	-2.28	Black Panther	-0.21	(0.239)	0.423	65,701	1.13	50.4	+3.08
Panel C: Support (13 heroes)								Human Torch	-0.28	(0.351)	0.468	55,560	0.96	54.1	+4.84
Mantis	6.76***	(0.181)	0.000	148,036	2.55	56.9	+6.74	Elsa Bloodstone	-0.39	(0.229)	0.108	89,462	1.54	50.1	+0.26
Rocket Raccoon	4.14***	(0.150)	0.000	198,679	3.43	52.5	+1.03	Psylocke	-0.58	(0.342)	0.106	46,440	0.80	50.3	+2.12
Ultron	3.59***	(0.297)	0.000	39,543	0.68	53.3	+3.17	Deadpool (Duelist)	-0.67***	(0.220)	0.003	57,041	0.98	51.5	+2.52
Adam Warlock	2.08***	(0.315)	0.000	40,028	0.69	51.5	+1.84	Namor	-0.82***	(0.243)	0.001	75,351	1.30	45.2	-6.75
Loki	1.18***	(0.268)	0.000	48,317	0.83	51.3	+0.66	Hawkeye	-1.22***	(0.341)	0.001	40,267	0.69	48.8	-1.03
Cloak & Dagger	-0.04	(0.170)	0.852	320,390	5.53	49.9	+0.42	The Punisher	-1.81***	(0.262)	0.000	75,297	1.30	46.2	-8.64
Gambit	-0.30	(0.210)	0.178	195,439	3.37	49.9	-0.34	Star-lord	-2.41***	(0.300)	0.000	51,580	0.89	49.7	-1.64
Invisible Woman	-1.38***	(0.151)	0.000	228,763	3.95	48.0	-4.01	Scarlet Witch	-2.49***	(0.414)	0.000	46,118	0.80	46.1	-3.43
Jeff The Land Shark	-1.75***	(0.233)	0.000	71,748	1.24	46.8	-3.27	Squirrel Girl	-3.78***	(0.350)	0.000	32,402	0.56	42.9	-12.83
Jubilee	-1.91***	(0.172)	0.000	290,670	5.02	50.5	+2.23	Cyclops	-4.04***	(0.195)	0.000	108,037	1.86	45.2	-3.61
Luna Snow	-3.01***	(0.167)	0.000	197,630	3.41	46.6	-4.24	Phoenix	-4.28***	(0.345)	0.000	45,826	0.79	44.0	-9.38
White Fox	-3.48***	(0.208)	0.000	99,670	1.72	46.1	-3.21								
Deadpool (Strategist)	-5.89***	(0.210)	0.000	59,610	1.03	45.5	-1.02								
Matches	477,483							Log-likelihood	-320,067.7						
Player–matches	5,729,796							Heroes	55						

Notes. All 55 heroes are listed. *APM* is in percentage points of win probability at a balanced match ($\partial p/\partial x = \beta/4$): the change from fielding this hero at match start in place of an average hero of the same role. Standard errors in parentheses are HC1; the 95% confidence interval is APM ± 1.96 S.E. Stars and q denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: * $q < 0.10$, ** $q < 0.05$, *** $q < 0.01$; 10 of the 55 are not distinguishable from their role average. *Starts* is the number of player-slots that began the match on this hero and *Pick %* its share of all starting slots; both reflect the crawl's per-hero leaderboard seeding, not population pick rates. *Raw WR* is the play-time-weighted raw win rate, an assumption-light benchmark that controls for nothing. *Play-time − Starting* (W1−W0) is how far play-time-weighted attribution inflates the estimate relative to starting-hero attribution, in points. Sample: 482,997 matches crawled, less 2,139 containing draws (`is_win` = 2), 19 with incomplete play time, and 3,356 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped. Appendix Table A3 repeats every hero under all three attribution rules.

Appendix

Table A1: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	within-role	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.801$	up from 0.671 before correction
Within-role reparameterisation	$\rho = 0.9999$	mean change 0.010 pp
Camp indexing	49.87% camp 0	absolute map side, no crawl selection
Temporal holdout (7 days)	log loss 0.6674	vs 0.6928 constant; slope 1.067

The placebo shuffles hero assignment against outcomes and refits; a placebo that failed would invalidate the specification. Camp-0 win rate varies 49.42%–52.52% across the 16 maps, which is why each map carries its own intercept. The holdout row summarises Figure 2.

Table A2: Out-of-sample fit as blocks of the model are added

Model	Parameters	Log-lik. (train)	Log loss (train)	Log loss (holdout)
Map intercepts only	16	−303,099	0.6929	0.6927
+ rank-score differential	17	−301,826	0.6900	0.6874
+ team-composition shapes	35	−299,416	0.6845	0.6819
+ hero contrasts	87	−294,223	0.6726	0.6698
+ team-up contrasts	187	−293,362	0.6706	0.6674

Each row adds one block to the model in the row above; the last row is the headline specification. Fitted on the 437,453 matches before the final 7 days, evaluated on the 40,030 matches of that week. A constant prediction at the training base rate gives a holdout log loss of 0.6928; the in-sample column shows how little of the improvement is overfitting.

Table A3: What happens to estimated hero strength when swaps affect attribution?

	more post-start information ⟶					*more post-start information* ⟶			
	Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)		Starting hero (W0)	Most-played hero (W2)	Play-time weighted (W1)	Difference (W1−W0)
Panel A: Tank (15 heroes)					*Panel B: Damage (27 heroes)*				
The Thing	5.46	4.94	7.21	+1.75	Black Cat	4.85	7.74	11.62	+6.77
Magneto	1.96	0.55	3.57	+1.60	Magik	3.47	7.94	14.57	+11.11
Angela	1.87	2.00	2.68	+0.81	Storm	3.09	6.57	10.27	+7.18
Peni Parker	1.51	10.18	7.41	+5.90	Daredevil	2.92	5.70	8.47	+5.55
Hulk	1.33	2.03	3.55	+2.21	Mister Fantastic	2.52	3.02	3.95	+1.43
Thor	1.29	0.73	0.65	-0.64	Iron Man	1.34	2.21	3.10	+1.76
The Hood	0.45	0.35	6.52	+6.08	Iron Fist	1.32	2.84	4.52	+3.19
Devil Dinosaur	-0.03	-0.56	-2.24	-2.21	Hela	1.14	1.04	1.22	+0.08
Groot	-0.67	-1.52	-3.79	-3.13	Winter Soldier	1.00	-0.07	0.17	-0.83
Venom	-0.93	-1.25	-1.35	-0.41	Wolverine	0.82	0.62	0.21	-0.61
Doctor Strange	-0.98	-2.55	-6.82	-5.85	Spider-man	0.58	1.48	2.90	+2.32
Captain America	-1.22	-0.51	-1.78	-0.57	Moon Knight	0.10	-2.01	-3.37	-3.47
Emma Frost	-1.77	-4.62	-6.68	-4.91	Black Widow	0.02	-0.96	-1.24	-1.26
Rogue	-3.41	-3.98	-1.78	+1.64	Blade	-0.20	0.28	1.09	+1.28
Deadpool (Vanguard)	-4.86	-5.79	-7.15	-2.28	Black Panther	-0.21	1.20	2.87	+3.08
Panel C: Support (13 heroes)					Human Torch	-0.28	1.74	4.56	+4.84
Mantis	6.76	9.88	13.50	+6.74	Elsa Bloodstone	-0.39	-0.46	-0.12	+0.26
Rocket Raccoon	4.14	4.50	5.18	+1.03	Psylocke	-0.58	0.08	1.54	+2.12
Ultron	3.59	6.92	6.76	+3.17	Deadpool (Duelist)	-0.67	0.48	1.85	+2.52
Adam Warlock	2.08	4.11	3.93	+1.84	Namor	-0.82	-3.87	-7.57	-6.75
Loki	1.18	2.14	1.84	+0.66	Hawkeye	-1.22	-1.31	-2.25	-1.03
Cloak & Dagger	-0.04	-1.84	0.38	+0.42	The Punisher	-1.81	-5.38	-10.45	-8.64
Gambit	-0.30	-2.07	-0.64	-0.34	Star-lord	-2.41	-2.64	-4.05	-1.64
Invisible Woman	-1.38	-3.32	-5.39	-4.01	Scarlet Witch	-2.49	-4.65	-5.91	-3.43
Jeff The Land Shark	-1.75	-3.04	-5.01	-3.27	Squirrel Girl	-3.78	-8.62	-16.61	-12.83
Jubilee	-1.91	-1.94	0.32	+2.23	Cyclops	-4.04	-5.45	-7.65	-3.61
Luna Snow	-3.01	-5.00	-7.25	-4.24	Phoenix	-4.28	-7.54	-13.67	-9.38
White Fox	-3.48	-4.73	-6.69	-3.21					
Deadpool (Strategist)	-5.89	-5.62	-6.91	-1.02					
Log-likelihood	−320,122	−305,869	−297,648		Cross-hero s.d.	2.60	4.27	6.38	
Rank corr. with starting hero	—	0.937	0.890						

Notes. Starting-hero attribution assigns the match to the hero picked first. Most-played attribution assigns it to the hero played longest. Play-time weighting divides attribution across every hero played according to time spent on each. Within-role estimates in percentage points, all three fitted on the same 477,483 matches. Effect magnitude rises monotonically with how much post-start information the rule uses, while the ranking is broadly preserved. The difference (W1−W0) is reported rather than a ratio because a ratio diverges for heroes whose starting-hero estimate is near zero. Higher log-likelihood here indicates worse identification, not a better model: play-time weighting predicts better precisely because its regressors encode what happened.

Table A4: Team-up synergies, all 100 pairs

Team-up	Premium	S.E.	Team-up	Premium	S.E.	Team-up	Premium	S.E.
Squirrel Girl × Iron Man	8.66	(0.965)	White Fox × Black Cat	1.77	(0.587)	Loki × Hela	0.45	(0.708)
Blade × Moon Knight	8.30	(0.851)	Black Panther × Storm	1.77	(0.996)	Hela × Namor	0.21	(0.552)
Rogue × Gambit	6.95	(0.472)	Hela × Venom	1.74	(0.552)	Gambit × Jubilee	0.21	(0.322)
Adam Warlock × Ultron	6.82	(0.768)	Thor × Hela	1.71	(0.552)	Mister Fantastic × The Thing	0.17	(0.773)
Iron Fist × White Fox	6.61	(0.498)	Venom × Phoenix	1.67	(0.596)	Luna Snow × Adam Warlock	0.16	(0.763)
Mantis × Star-lord	4.27	(0.503)	Mister Fantastic × Rocket Raccoon	1.62	(0.591)	Devil Dinosaur × The Punisher	0.10	(0.700)
The Punisher × Rocket Raccoon	4.13	(0.422)	Blade × Captain America	1.62	(0.981)	Namor × Luna Snow	0.09	(0.483)
Storm × Jeff The Land Shark	3.86	(0.667)	The Punisher × Daredevil	1.58	(0.850)	Spider-man × Venom	0.06	(0.424)
Human Torch × Jubilee	3.71	(0.469)	Hawkeye × Cloak & Dagger	1.58	(0.542)	Groot × Jeff The Land Shark	-0.02	(0.819)
Loki × Mantis	3.53	(0.667)	Ultron × Iron Man	1.54	(0.864)	Psylocke × Cloak & Dagger	-0.03	(0.514)
Doctor Strange × Hulk	3.47	(0.603)	Emma Frost × Mantis	1.50	(0.363)	Venom × Blade	-0.04	(0.750)
Magik × The Hood	3.36	(0.295)	Daredevil × Black Widow	1.48	(0.728)	Captain America × Thor	-0.06	(1.079)
Adam Warlock × Storm	3.28	(1.149)	Wolverine × Gambit	1.47	(0.581)	Moon Knight × Elsa Bloodstone	-0.06	(0.940)
Luna Snow × White Fox	3.19	(0.447)	Magik × Doctor Strange	1.45	(0.389)	Winter Soldier × The Punisher	-0.07	(0.538)
Peni Parker × Rocket Raccoon	3.05	(0.543)	Jeff The Land Shark × Venom	1.41	(0.600)	Magneto × Emma Frost	-0.12	(0.313)
Cyclops × Phoenix	2.79	(0.641)	Angela × Star-lord	1.39	(1.006)	Black Cat × Black Panther	-0.25	(1.018)
Mantis × Adam Warlock	2.76	(0.831)	Angela × Loki	1.35	(0.957)	Jubilee × Blade	-0.41	(0.519)
Cloak & Dagger × The Hood	2.60	(0.233)	Invisible Woman × Human Torch	1.32	(0.507)	Invisible Woman × Mister Fantastic	-0.44	(0.614)
Doctor Strange × Invisible Woman	2.58	(0.326)	Gambit × Magneto	1.31	(0.277)	The Thing × Human Torch	-0.45	(0.548)
Scarlet Witch × Jubilee	2.47	(0.510)	Star-lord × Groot	1.24	(0.784)	Psylocke × Emma Frost	-0.49	(0.615)
Jubilee × The Hood	2.46	(0.236)	Elsa Bloodstone × Devil Dinosaur	1.21	(0.583)	Magneto × Scarlet Witch	-0.55	(0.511)
Black Widow × Phoenix	2.44	(0.859)	The Thing × Invisible Woman	1.20	(0.358)	Rocket Raccoon × Groot	-0.55	(0.518)
Hulk × Captain America	2.39	(0.998)	Rogue × Magneto	1.07	(0.501)	Thor × Angela	-0.81	(1.007)
Hawkeye × Psylocke	2.35	(0.907)	Cloak & Dagger × Luna Snow	1.07	(0.318)	Iron Man × Hulk	-0.86	(0.796)
Squirrel Girl × Spider-man	2.17	(0.959)	Captain America × Winter Soldier	0.98	(0.672)	Cyclops × Gambit	-1.00	(0.411)
Ultron × Peni Parker	2.11	(1.072)	Namor × Hulk	0.86	(0.585)	Storm × Thor	-1.17	(0.733)
Spider-man × Peni Parker	2.02	(0.714)	Devil Dinosaur × Jeff The Land Shark	0.83	(0.679)	Hulk × Wolverine	-1.39	(0.805)
Winter Soldier × Elsa Bloodstone	2.00	(0.442)	Phoenix × Rogue	0.79	(1.066)	Black Widow × Hawkeye	-1.54	(1.126)
Phoenix × Hela	1.98	(0.808)	Iron Fist × The Thing	0.73	(0.490)	Daredevil × Iron Fist	-1.80	(0.741)
Moon Knight × Cloak & Dagger	1.97	(0.465)	Emma Frost × Luna Snow	0.70	(0.332)	Human Torch × Storm	-1.80	(1.057)
Star-lord × Adam Warlock	1.95	(1.190)	Wolverine × Cyclops	0.68	(0.881)	Black Panther × Magik	-2.09	(0.666)
Iron Man × Thor	1.87	(0.688)	White Fox × Psylocke	0.61	(0.778)	Black Cat × Spider-man	-2.24	(0.627)
Scarlet Witch × Doctor Strange	1.85	(0.628)	Peni Parker × Black Panther	0.52	(1.041)			
Groot × Mantis	1.84	(0.535)	Rocket Raccoon × Squirrel Girl	0.50	(0.726)			

Notes. All 100 team-ups are listed in descending order of premium, reading down each column in turn. Each is labelled by its member heroes, anchor first. The premium is what a pair earns *beyond* what its two heroes contribute individually, in percentage points, under starting-hero attribution; HC1 standard errors in parentheses. Four further team-ups are absent because they are defined against hero id 1057, base "Deadpool", whose plays are all recorded under his three role variants, leaving those columns structurally empty.